

Cell Specialisation

Answer sheet · 24 marks total

Section A — Specialised animal cells

- A1.** Long tail (1), lets it swim to the egg (1). *[or: many mitochondria (1), for energy to swim (1); or: acrosome (1), digests through the egg membrane (1)]*
- A2.** Long axon (1), carries signals over long distances (1). *[or: branched endings (1), connects to many other cells (1); or: insulating layer (1), speeds up the signal (1)]*
- A3.** Long fibre that contracts (1), shortening causes movement (1). *[or: many mitochondria (1), energy for contraction (1)]*

Section B — Specialised plant cells

- B1.** Long hair-like extension (1), gives a large surface area for absorbing water/mineral ions (1).
- B2.** Hollow tube with no contents (1), lets water flow through easily (1). *[or: lignin-thickened walls (1), gives strength/support (1)]*
- B3.** Sieve tube with pores (1), lets sugars flow through (1). *[or: companion cell alongside (1), provides energy for active transport of sugars (1)]*

Section C — Explain in full

- C1.** Long tail lets it swim to the egg (1); many mitochondria release energy for swimming (1); acrosome contains enzymes that digest through the egg membrane (1).
- C2.** Hollow tube with no cell contents lets water flow through easily (1); walls are thickened with lignin, making it strong (1); this lets it support the plant as well as transport water (1).

Section D — Unfamiliar cells

- D1.** The cilia beat back and forth (1); this sweeps mucus (and trapped dust/bacteria) along and out of the airway (1); this helps protect the lungs from infection/damage (1).
- D2.** Secreting hormones requires energy (e.g. for active transport) (1); mitochondria are the site of respiration, which releases energy (1); more mitochondria means more energy is available for secretion (1).

Marking note: accept any scientifically correct feature/function pair in Sections A and B, even if different from the example shown — one correctly linked feature is worth both marks. In Section D, accept any logically sound application of the 3-step method.